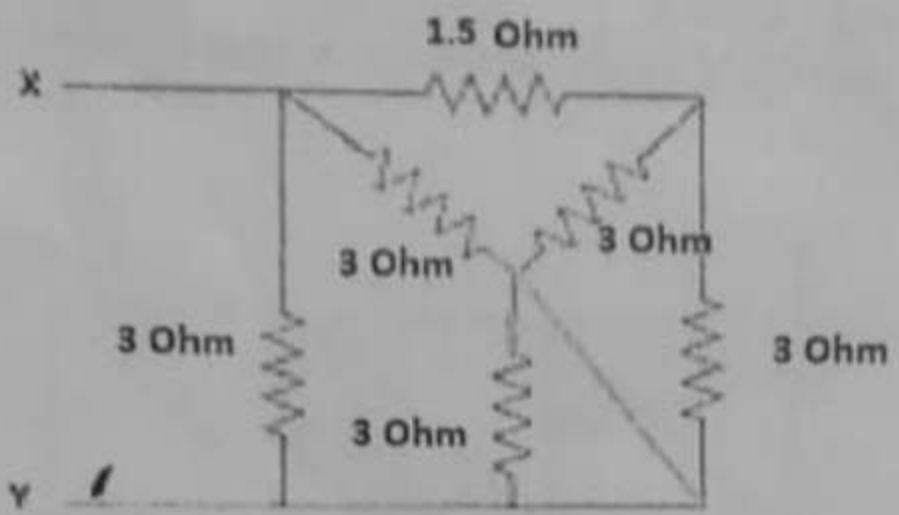
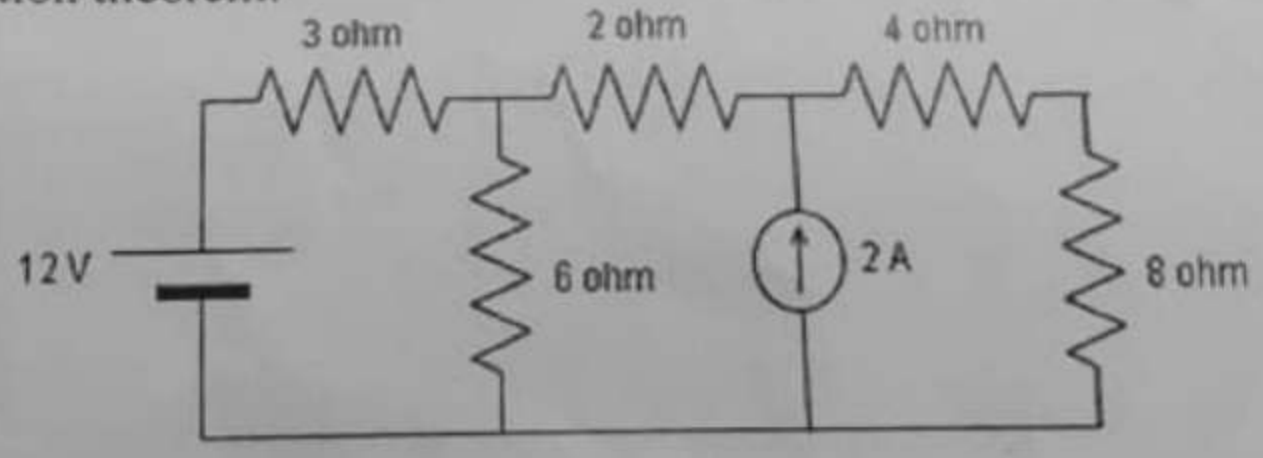
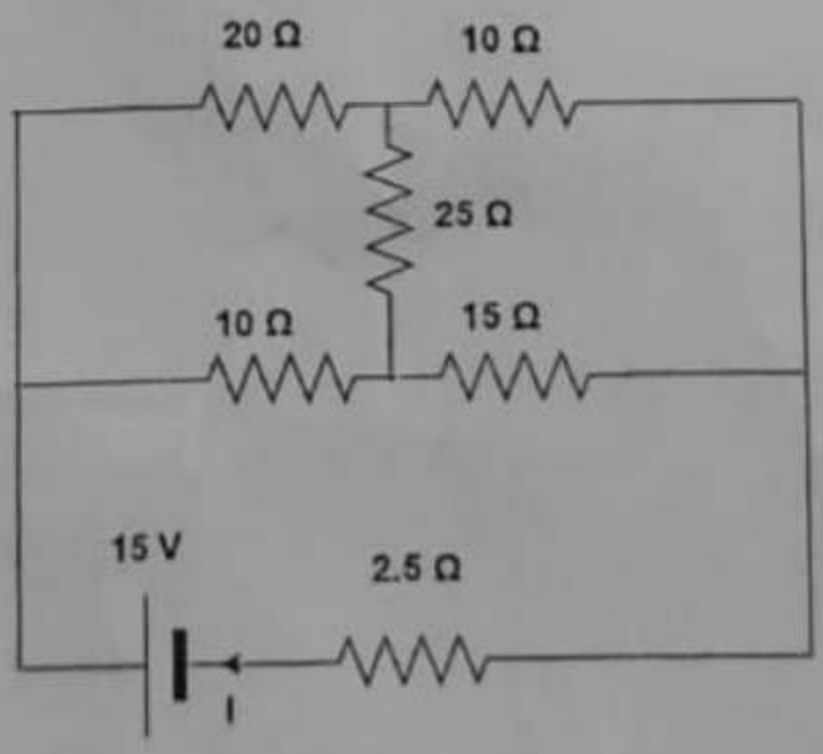
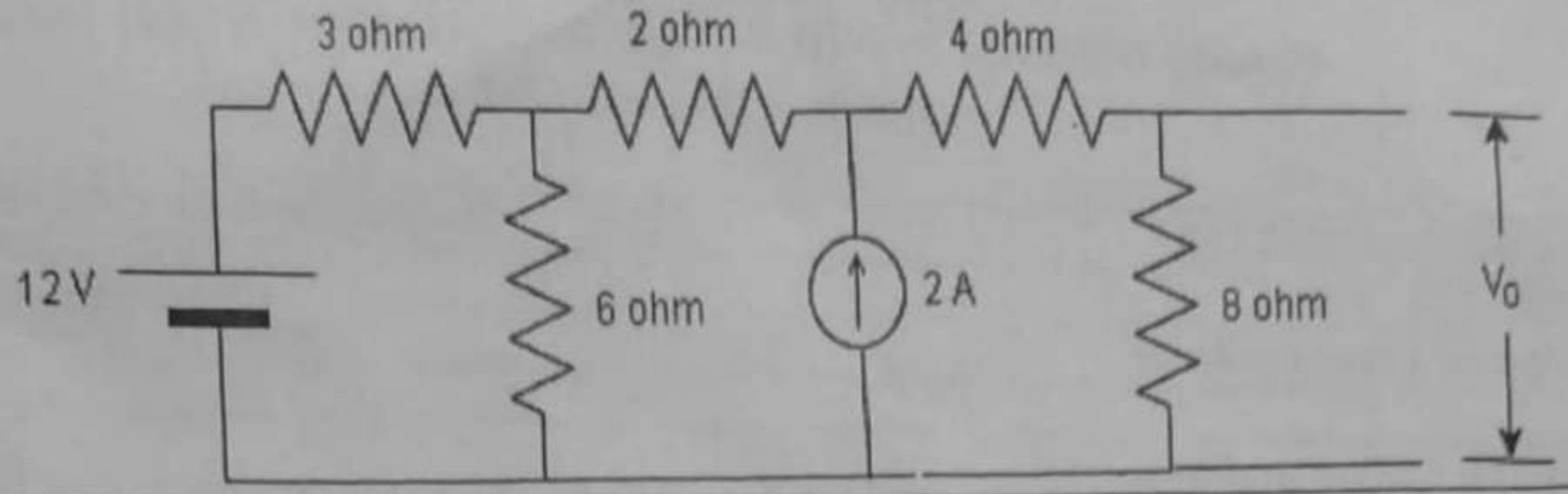
MID-TERM EXAMINATION
(Course Name: B. Tech (IT/MAE/DMAM) (Semester: 2nd) - Regular
(March, 2024) OFF LINE mode

Subject Code: BEC 110	**Subject: Basic Electrical Engineering**
Time: 1 ½Hours	**Maximum Marks:** 30
Note: Q. 1 is compulsory.	

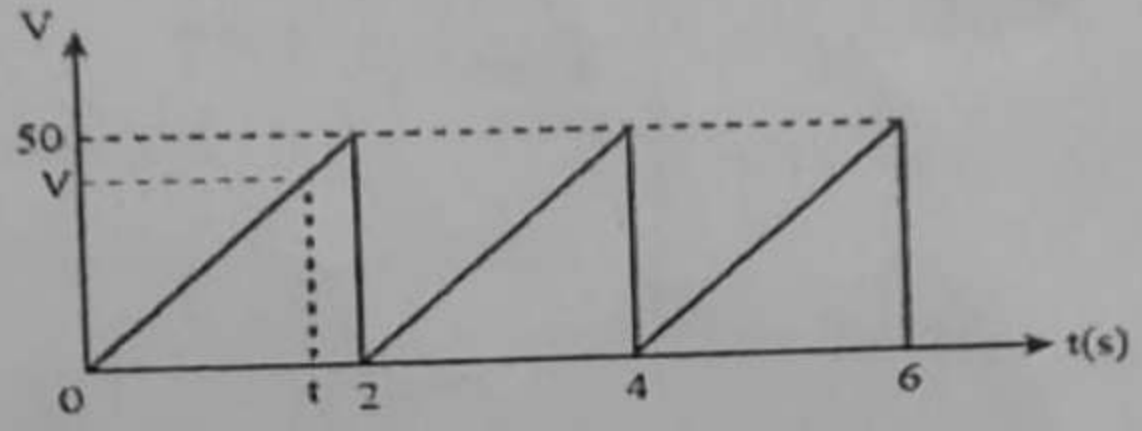
Q1	**Q. 1 is compulsory.**		**(2.5*4)**	
	(a) Find the equivalent resistance between X and Y. 		UNIT-1	
	(b) State and Prove Maximum Power Transfer Theorem. Derive the expression of Maximum Power transferred.		UNIT-1	
	(c) Two impedances 5+j8 and 3-j8 are connected in parallel. Find the equivalent impedance of the combination.		UNIT-2	
	(d) Draw power triangle. Explain true power, reactive power and apparent power?		UNIT-2	
Q2	**(Attempt any Two Parts)** **UNIT-1**			**(5,5)**
	(a) Define Superposition Theorem. Find current in 8 Ohm resistance using Superposition theorem. 			
	(b) Determine the total current I in the given circuit. 			

(c) Derive the Thevenin's equivalent circuit of the figure given below.



Q3	(Attempt any Two Parts) UNIT-2	(5,5)

(a) Find the average and RMS value of the waveform shown below. Do find peak factor and form factor.



(b) A pure resistance of 50 Ω is connected is in series with a pure capacitor of 100 μF. The series combination is connected across 100 V, 50 Hz supply. Find (i) Impedance (ii) Current (iii) Power Factor (iv) Voltage across R and C (v) Draw phasor diagram

(c) A series RLC circuit is connected across an ac voltage source ($v=V_m \sin\omega t$). Draw the phasor diagram and derive the following: (i) Total impedance (ii) Expression of current (iii) Resonance frequency and (iv) Quality factor.